ASSIGNMENT 4

Q1: E-Commerce Product Manager

Instructions: Save JavaScript-only solutions as `.js` files and run with Node.js. For form questions, save the full code as `.html` and open it in a browser. No frameworks are used.

Solution

```
class Product {
  constructor(id, name, price, category) {
    Object.assign(this, { id, name, price, category });
  }
  applyDiscount(percent) {
    this.price -= this.price * percent / 100;
  }
  getDetails() {
    return `${this.name} (${this.category}) - ■${this.price.toFixed(2)}`;
  }
}

const products = [
  new Product(1, "Keyboard", 1500, "Accessories"),
  new Product(2, "Mouse", 800, "Accessories"),
  new Product(3, "Monitor", 12000, "Electronics")
];

products[0].applyDiscount(10);
products.forEach(product => console.log(product.getDetails()));
console.log("Products above ■1000:");
products.filter(product => product.price > 1000)
  .forEach(product => console.log(product.getDetails()));
```

Expected result / testing

Run with Node.js. Keyboard becomes ■1350 after discount; the filter lists Keyboard and Monitor.